

REWEAR-FUSED: Six Blind-Spot Audit — Verified Defensive Brief for Hackathon Judging

TL;DR

- Three of the six blind spots are *near-fatal as currently framed* and one is *fatal*: (1) **MDA/TDA toxicity** is fatal if the team designs around MDI/TDI hard segments for an infant-apparel use case — but it becomes a **non-issue** if Engine A is constrained to aliphatic isocyanates (HDI, H12MDI, IPDI), which is the correct pivot; (2) **UMG-SP2's pH-10 optimum** is real and incompatible with cotton/wool recovery, so the team **MUST** present Engine B as engineering a near-neutral-pH variant rather than using wild-type UMG-SP2 verbatim; (3) the **mechanics-vs-cleavability** trade-off is real but a published design window does exist (PCL/polycarbonate soft segments with hydrolyzable bonds in the soft phase).
 - The two **most easily neutralized** blind spots are #4 (manufacturing CapEx — already proven by Hyosung Creora bio-based and LYCRA+QIRA precedents, with a credible 5-10 year licensing pathway to Hyosung/LYCRA/Asahi Kasei) and #6 (Engine A reframe — the team should openly say "AI-assisted constraint-guided virtual screening of PU-urea chemical space using polyBERT-style property predictors," not "de novo generative design," because polyBART and polyBERT do not handle segmented block copolymers).
 - The **most dangerous unresearched gap** the team has not flagged is *microplastic shedding during use phase and end-of-life leaching of any unreacted aliphatic diamine into wash water* — a sustainability judge will probe this, and the team should pre-empt it with the EU PFAS/microplastics regulatory framing.
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BLIND SPOT 1 — Toxicity & Infant-Skin Safety of Degradation Products

Verified facts

MDA (4,4'-methylenedianiline, CAS 101-77-9) is the verified hydrolysis product of MDI-based polyurethane hard segments. The hydrolysis of urethane bonds in MDI-based PUs releases MDA directly; this is documented in medical-device, dialyzer, and PU-glycolysis literature (Shintani & Nakamura, *Fresenius Z. Anal. Chem.* 1989, 333, 637–640, DOI 10.1007/BF00496937; pharmacokinetic modeling, PubMed PMID 10813768). The 2023 Branson urethanase paper explicitly demonstrates that enzymatic hydrolysis of TDI-based polyether-PU yields **toluenediamine (TDA)** as the aromatic diamine end-product (Branson et al., *Angew. Chem. Int. Ed.* 2023, 62, e202216220, DOI 10.1002/anie.202216220). (ResearchGate) (RSC Publishing)

Carcinogenicity status:

- **IARC: MDA is classified Group 2B (possibly carcinogenic to humans).** US EPA Hazard Summary, <https://www.epa.gov/sites/default/files/2016-09/documents/4-4-methylenedianiline.pdf>; ScienceDirect Encyclopedia of Toxicology (2024).
- MDA is **mutagenic and carcinogenic in animals**, a hepatotoxin, contact and respiratory allergen (PMID 10813768).
- California EPA oral cancer slope factor: $1.6 \text{ (mg/kg/d)}^{-1}$.

Regulatory limits applicable to infant textiles (Carter's target market):

- **REACH Annex XVII Entry 43, Appendix 8:** 22 carcinogenic aromatic amines (the historical "azo dye amines" list including 4,4'-MDA and 2,4-/2,6-TDA) restricted to **30 mg/kg (0.003 % by weight)** in dyed textile/leather articles with prolonged skin contact. Enforced via EN ISO 14362-1:2017 (textiles) and EN ISO 17234-1:2015 (leather). Source: SATRA, <https://www.satra.com/spotlight/article.php?id=594>. (SATRA) (Intertek)
- **REACH Annex XVII Entry 72** (Regulation (EU) 2018/1513, in force 1 Nov 2020): 33 CMR substances restricted in clothing, textiles and footwear, including specific aromatic amines and carcinogenic dyes. European Commission, https://single-market-economy.ec.europa.eu/sectors/chemicals/reach/restrictions_en. (European Commission)
- **OEKO-TEX Standard 100 Class I (infants ≤36 months):** applies the strictest limits across >1,000 substances; carcinogenic aromatic amines from azo dyes **must not be detectable above 20 mg/kg in any of the four product classes** (OEKO-TEX Factsheet, 2023, https://www.oeko-tex.com/fileadmin/user_upload/Marketing_Materialien/STANDARD_100/Factsheet/STANDARD_100/OEKO-TEX_STANDARD_100_Factsheet_EN.pdf). Class I is the *only* class Carter's Little Planet/baby line can credibly market under. (Q for Quinn)
- **CPSIA (16 CFR Parts 1307, 1610, etc.):** regulates lead (90/100 ppm), phthalates (8 phthalates ≤0.1 %), flammability, but does NOT directly regulate aromatic amines from PU hydrolysis. However, **California Proposition 65 lists MDA** as a chemical known to cause cancer (Cal EPA OEHHA listing); MDA on Prop 65 requires warning labels for any consumer product in California exceeding the No Significant Risk Level. (Trustrace)

Aliphatic isocyanate route eliminates the carcinogenic-amine problem

Verified design strategy: Aliphatic diisocyanates **HDI (hexamethylene)**, **H12MDI (hydrogenated MDI)**, and **IPDI (isophorone)** are commercial, well-characterized, and on hydrolysis release *aliphatic* diamines (1,6-hexamethylenediamine, 4,4'-diaminodicyclohexylmethane, isophoronediamine) — **none of which are IARC carcinogens and none are on the REACH Annex XVII Appendix 8 carcinogenic-amine list.** The aliphatic-amine

route is the standard design approach for biocompatible/biodegradable PU elastomers (Tri-iso, <https://www.tri-iso.com/polyurethane-and-polyurea-aliphatic-isocyanates.html>; ULProspector, <https://www.ulprospector.com/knowledge/5144/pc-polyisocyanates-deep-dive/>; PMC9002640 — non-toxic biodegradable PUs with aliphatic-isocyanate hard segments). Aliphatic PU is the dominant choice in biomedical-grade resorbable PU (ScienceDirect, Aromatic Isocyanate overview). Drawbacks: aliphatic isocyanates react ~10× slower than aromatic, require strong catalysts (dibutyltin dilaurate), and are more expensive — but these are processing concerns, not safety blockers. (PCI Magazine)

Verdict: MANAGEABLE if Engine A is constrained to aliphatic isocyanates. FATAL if the team specs MDI/TDI hard segments.

Defensible answer to a hostile judge

"Our Engine A design space is *constrained* to aliphatic isocyanates — specifically H12MDI and HDI — for the explicit reason you're raising. We will not generate any candidate that releases MDA or TDA on enzymatic cleavage. The degradation products are aliphatic diamines (e.g., 4,4'-diaminodicyclohexylmethane from H12MDI), none of which appear on the IARC 2B list, REACH Annex XVII Appendix 8 carcinogenic-amine list, or OEKO-TEX Standard 100 Class I restricted substances list. Pre-commercial validation would require **EN ISO 14362-1:2017 migration testing for the 22 aromatic amines, ISO 10993-10 (skin sensitization) and ISO 10993-5 (cytotoxicity) on extractables, and an OEKO-TEX Class I certification** on the finished yarn before any infant-apparel placement."

BLIND SPOT 2 — pH Compatibility with Cotton/Wool

Verified facts

UMG-SP2 pH optimum is verifiably pH 10 with a reference activity-assay condition of pH 8.0, 30 °C, 50 mM Na-phosphate. The primary source is Branson et al. (*Angew. Chem. Int. Ed.* 2023, 62, e202216220, DOI 10.1002/anie.202216220), which states verbatim (translated from the German edition): "In the range from pH 4 to pH 11, all enzymes were most active at pH 10. UMG-SP-1 and UMG-SP-2 were most active at 70 °C, the highest temperature we examined." The standard PU-degradation assay in Li et al. 2025 (*Adv. Sci.* 12, 2416019, DOI 10.1002/advs.202416019) uses 100 mM glycine-NaOH pH 9.0, 30 °C. The Świderek et al. JACS 2025 QM/MM study (DOI 10.1021/jacs.5c13147, PMC12636025) explicitly confirms the mechanistic basis: "constant-pH nonequilibrium molecular dynamics and Monte Carlo (neMD/MC) simulations provided the full titration curve of active site Lys, **explaining the need**

for alkaline media for the enzyme to be active" and reports "UMG-SP2 hydrolyzes pNC more efficiently at alkaline pH (pH = 10)." The red-team claim that UMG-SP2 requires alkaline pH (~8.5–10.5) is CORRECT — the catalytic Ser-Sercis-Lys triad requires the lysine ϵ -amine (pKa ~10.5 in solution) to be predominantly deprotonated, which only occurs above pH ~9. No verbatim numeric residual-activity figure at pH 7 appears in the main text of Branson 2023, Bayer 2024, or Li 2025; it must be read from the supplementary activity-pH curve.

ResearchGate + 4

Effect of alkaline pH on companion fibers:

- **Cotton (cellulose):** Mercerization-grade alkaline damage requires very high NaOH (~18–25 % w/w, ~4–6 M, pH >>13) at room temperature (ScienceDirect Mercerisation overview; causticsodaco.com). However, alkaline conditions at elevated temperature (≥ 60 °C) for prolonged times cause "tendering" — slow base-catalyzed chain scission of cellulose (PetroNaft caustic-soda guide). Reactive-dye fixation is conducted routinely at pH 10.5–12.0 without destroying cotton, so a brief pH-9–10 enzymatic bath at 30 °C is **compatible with cotton recovery**.
- **Wool (keratin):** Keratin is significantly more pH-sensitive. The classic Diego & Silver study (J. Invest. Dermatol., historical): "**The first reaction is one of peptisation up to pH 9 (swelling of the keratin complex—'quellung'). Above pH 9.2, and depending upon the concentration, time of contact, and temperature, rapid hydrolysis occurs.**" Modern hair literature confirms: "Alkali hydrolysis of peptide and disulfide bonds above pH 10 weakens the hair further and causes hair damage" (Malinauskyte, *Biopolymers* 2020, DOI 10.1002/bip.23401). Wool keratin hydrolysis at pH 12.5 / 98 °C achieves 100 % solubilization (Wiley CRFSFS 2023, DOI 10.1111/1541-4337.13087). (Wiley Online Library)

Conclusion: At pH 9.0–10.0 and 30 °C (Li 2025 reaction conditions), cotton survives, but wool sustains measurable disulfide-bond and amide-bond damage. Carter's apparel is overwhelmingly cotton (organic cotton, cotton/poly blends) with negligible wool, so for the specific Carter's use case the cotton-compatibility is the relevant constraint.

Co-design freedom for near-neutral pH

The team's de novo design freedom is *real* — the pH optimum of a serine hydrolase is governed by the pKa of the catalytic base. Replacing the Lys-acting-as-base with a **histidine** (pKa ~6.0–7.0) — i.e., engineering a classical Ser-His-Asp catalytic triad instead of the amidase-signature Ser-Sercis-Lys triad — would shift the pH optimum to near-neutral. This is mechanistically supported by the existence of acid/neutral-active urethanases: e.g., the engineered *Candida parapsilosis* urethanase mutant (cpUH G246A/N194V) shows kcat/Km maximum at pH 4.5 (ScienceDirect, <https://www.sciencedirect.com/science/article/pii/S0147651324004111>). A 2025

esterase-superfamily urethanase paper (PMC12456375) reports variants with pH optima of 7.0 (MthABH), 8.0 (TflABH), 9.5 (OspAmd). So the catalytic chemistry is not inherently restricted to alkaline pH — UMG-SP2 specifically needs it because of its Lys91-based mechanism. (nih)

Verdict: MANAGEABLE — but requires explicit framing that Engine B engineers a neutral-pH variant.

Defensible answer

"The wild-type UMG-SP2 pH-10 optimum is exactly why we are not using wild-type UMG-SP2. The Ser-Sercis-Lys triad requires alkaline pH to keep Lys91 deprotonated (Świderek 2025, JACS, DOI 10.1021/jacs.5c13147; Branson 2023 verbatim: 'all enzymes were most active at pH 10'). Engine B's de novo design freedom lets us *re-engineer* the catalytic machinery — most simply by replacing the catalytic lysine with a histidine to recover a Ser-His-Asp-like triad with a near-neutral pH optimum, as exists in serine proteases, lipases, and cutinases. Acid-/neutral-active carbamate hydrolases already exist (cpUH G246A/N194V active at pH 4.5; MthABH pH-7.0 optimum, 2025). Our target operating window for the bath is **pH 7.0–8.5 at 30–40 °C**, which is compatible with cotton recovery (mercerization requires >>4 M NaOH at pH >>13) and which causes negligible damage to keratin (the pH-9.2 threshold for wool peptide hydrolysis is not crossed)."

BLIND SPOT 3 — Cyclic Fatigue, Hysteresis, Wash Durability

Verified facts

Commercial spandex performance is defined by four properties beyond Tg/elongation-at-break:

- **Elastic recovery:** premium LYCRA®-grade ≥95–97 % recovery after maximum elongation; generic spandex 85–90 % (Hohenstein Institute 2023; Textile Testing Services UK 2022; Szoneierfabrics). (Szoneierfabrics)
- **Cyclic fatigue resistance:** premium LYCRA retains **95 % elasticity after 200,000 cycles**; generic spandex loses 15–20 % under identical conditions. (Szoneierfabrics)
- **Hysteresis:** low hysteresis = high energy return; defined by ASTM D2594 (growth/recovery) and ISO 20932-1:2018 (extension-recovery under defined load).
- **Wash-durability:** spandex is degraded by chlorine, heat (>40 °C), UV, and prolonged sunlight; typical garment lifespan 1–3 years of heavy use (Apex Fashion Lab, <https://www.apexfashionlab.com/glossary/spandex-lycra>). Sweat, body oils, detergents are acceptable.

Design tension is real and documented

Multiple peer-reviewed studies confirm that hard-segment crystallinity and hydrogen bonding are precisely what provide elastic recovery and low hysteresis:

- **Bonart-style work** (Journal of Macromolecular Science 1983, DOI 10.1080/00222348308245754): hysteresis decreases when hard-segment crystallinity is *disrupted* by mixed chain extenders that prevent lamellar growth — but at the cost of tensile strength.
- **DBDI vs MDI study** (Academia.edu PDF 69237008): "Hysteresis was increased by increasing hard phase crystallinity" but tensile modulus rose to 320 MPa for semicrystalline materials — confirming the trade-off: crystallinity gives strength and modulus but increases first-cycle hysteresis; reducing crystallinity reduces both hysteresis AND strength.
- **Qin et al., Adv. Sci. 2024** (DOI 10.1002/advs.202400255): PCL-based PU-urea elastomers with smaller hard-segment clusters showed *lower* residual strain (3 % vs 8 %) and excellent fatigue resistance over 100 cycles, achieved by careful chain-extender (DD vs AD) selection rather than crystallinity disruption.
- US Patent 9,644,289 (LYCRA-style polyurethaneurea elastic fiber): explicitly claims that a narrow hard-segment-content window gives "high recovery stress, low hysteresis loss, and high heat resistance" — confirming the team must hit a narrow design window, not just minimize crystallinity.

The fundamental design tension is REAL: disrupting hard-segment crystallinity for enzymatic accessibility directly attacks the physical-crosslink network that provides spandex's elastic recovery and fatigue resistance.

ML predictability of cyclic fatigue and elastic recovery

- Current polymer-property ML (polyBERT, PolyID, ADEPT-PolyGraphMT) is trained primarily on Tg, Tm, density, dielectric constant, Young's modulus — **NOT on cyclic fatigue, elastic recovery, or hysteresis.**
- The Springer 2022 ML study on PU elastomer mechanical properties (DOI 10.1007/s10118-022-2838-6) reports R^2 0.85–0.91 on Young's modulus, tensile strength, and elongation at break from a *curated* PUE dataset — but does NOT cover cyclic fatigue, recovery, or hysteresis.
- **No published ML model reliably predicts cyclic-fatigue retention or hysteresis from structure;** these require wet-lab measurement.

Verdict: MANAGEABLE design tension; ML limitation must be honestly conceded.

Defensible answer

"You are right that the same hard-segment crystallinity we disrupt for enzymatic accessibility is what provides elastic recovery — this is documented in the Bonart-tradition literature and the Qin *Adv. Sci.* 2024 study. We are not claiming to predict cyclic fatigue or hysteresis in silico; current polymer property-prediction models (polyBERT, PolyID) cover Tg, Tm, modulus, and elongation at break but not fatigue. Our reframed Engine A scores candidates on *predictable* properties (Tg, modulus, elongation) and on an *accessibility metric* (hard-segment H-bond density), then emits the top candidates for **wet-lab validation of recovery (ISO 20932-1:2018), 200,000-cycle fatigue (ASTM D2594), and recovery after laundering (AATCC 135)**. The realistic design window is in the **30–35 wt% hard-segment range with mixed-component hard segments** (Bonart-style) and **PCL or polycarbonate soft segments** that place the cleavable bond in the amorphous phase — not in the load-bearing crystalline domains."

BLIND SPOT 4 — Manufacturing CapEx and Drop-In Compatibility

Verified facts

Commercial spandex spinning:

- Solution dry-spinning held **more than 94 % of global spandex market share in 2021** (Straits Research). Mordor Intelligence (data as of January 2026) reports the updated share as **91.78 % in 2025**, reflecting modest loss to alternative methods. Wet spinning <5 %; melt/reaction spinning niche. (Straits Research + 2)
- Dry-spinning plant CapEx: **US\$40–50 million for a 3,000–5,000 ton/year plant at 40-denier average** (Fibre2Fashion, <https://www.fibre2fashion.com/industry-article/1/a-new-process-of-producing-spandex-yarns>). Indorama Synthetics announced a **US\$82 million** plant modernization/expansion in 2021.
- Dry-spinning chemistry: macroglycol (PTMEG) + diisocyanate (MDI) → prepolymer dissolved in DMAc → chain-extended with diamine → extruded through spinneret into heated cell with N₂/solvent gas (Procurement Resource, Textile-Craft).
- Major players: Hyosung TNC (~32 % global market share in 2020), LYCRA Company (formerly INVISTA), Asahi Kasei, Indorama. **In 2023 China alone produced 939,000 tons of spandex (+19.5 % YoY) with installed capacity of 1.2395 million tons/year at end-2023** (CCFGroup, reported by market.us, Dec 2024). (Market.us)

Novel-fiber commercialization timelines (precedents):

- **Hyosung Creora bio-based:** announced and commercialized August 2022, USDA/SGS certified, 30 % bio-content corn-BDO replacing fossil BDO. Per Hyosung TNC Business Wire (Aug 16, 2022): "Hyosung TNC has made research and development efforts for more than a year, and successfully commercialized bio-based spandex for the first time in the world." LCA: "**reduces water use by 39 % and carbon dioxide emissions by 23 % compared to conventional spandex products.**" Drop-in: same dry-spinning equipment. Textileworld + 2
- **LYCRA Company × QIRA® (Qore JV of Cargill+HELM):** announced September 2022, letter of intent with Dairen Chemical (DCC) May 2024, commercial launch early 2025. Per the official LYCRA/DCC Business Wire press release (May 9, 2024): "**Available in early 2025, patented bio-derived LYCRA® fiber made with QIRA® will be the world's first renewable spandex available on a large scale.**" Total R&D-to-market: ~3 years. 70 % bio-derived content via bio-PTMEG; up to 44 % CO₂ reduction. LYCRA explicitly states: "Stretch, recovery, and other parameters will deliver equivalent functionality to current product" — drop-in compatible. WTIN + 4
- **Asahi Kasei ROICA™ V550:** world-first Cradle-to-Cradle Gold-certified, partially biodegradable per ISO 14855-1, available 22-1870 dtex. Per Just-Style (Sept 14, 2018): "The Roica V550 stretch yarn has held Cradle-to-Cradle Certification since 2016." The first major commercial brand partnership (C&A's world-first C2C Certified jeans) launched September 2018 — approximately **2 years from certification to first brand launch.** Asahi Kasei + 3
- **The critical insight: ALL recent "sustainable spandex" innovations preserve dry-spinning compatibility by substituting a single feedstock (BDO → bio-BDO; or chemistry → cleavable chemistry) rather than redesigning the spinning process.**

Drop-in feasibility for Engine A: If the team's novel PU-urea is (a) soluble in DMAc, (b) viscosity range compatible with the spinneret (~5,000–100,000 cP at process temperature), (c) chain-extends with a diamine in solution, and (d) cures to a fiber in the heated spinning cell, **it is drop-in compatible.** The aliphatic-isocyanate switch (Blind Spot 1) means reaction kinetics are ~10× slower than aromatic — this is the **only meaningful processing change**, and is managed with tin/bismuth catalysts already used commercially in coatings.

Verdict: MANAGEABLE — credible licensing path to incumbent.

Defensible answer to a Cox Cleantech Residency judge

"Our commercialization story is not 'build a new fiber plant.' It is: develop the novel polymer chemistry to TRL 4, then license to an incumbent (Hyosung, LYCRA, Asahi Kasei) who already operates dry-spinning lines. The benchmark precedents are **Hyosung Creora bio-based (R&D-to-market ~1 year, 23 % CO₂ reduction)**, **LYCRA × QIRA (~3 years, early 2025 commercial**

launch), and ROICA V550 (~2 years from C2C certification 2016 to first brand launch 2018).

All three preserve the existing US\$40–50M dry-spinning asset base by substituting a single feedstock or chemistry block. Our fiber is designed to be drop-in to solution dry-spinning in DMAc — same prepolymer/chain-extension/spinneret workflow — with the only delta being an aliphatic-isocyanate-compatible catalyst (commercial). For Carter's specifically, the realistic adoption pathway is a 2028–2030 Hyosung Creora-class licensing deal at small percentage of garment composition (2 % spandex, Hyosung's denim model), with Carter's piloting in the Carter's Little Planet sub-line first."

BLIND SPOT 5 — Mechanics-vs-Cleavability Trade-Off Curve

Verified facts: a published design window DOES exist

- **Polycaprolactone (PCL) soft-segment biodegradable TPUs** are the textbook strategy. Gantrade industry note: "Polycaprolactone-based polyurethanes stand-alone in meeting a critical set of criteria for biodegradable PU elastomeric materials: open to hydrolysis, non-toxic degradation products, retained mechanical and thermal properties, TPU-processable." (<https://www.gantrade.com/blog/biodegradable-polycaprolactone-urethane-elastomers>).
- **Specific quantitative example:** BTPU with 60 % bio-polyester soft block — tensile strength **30 MPa**, elongation at break **>800 %**, **36.7 % mass loss in 14 d lipase solution at 80 % BPD content** (Hu et al., *Polymers* 2019, 11, 1121, <https://www.mdpi.com/2073-4360/11/7/1121>).
- **PCL polyol-based TPEs** (Sustainable PCL TPEs, PMC10421468): **39.7 MPa ultimate strength, 47.6 % biodegradability** — vs 23.4 MPa and 24.3 % for PTMEG-control TPEs. Crystallinity in the PCL soft segment is what gives both strength AND degradability.
- **Xu et al. *SusMat* 2026** (DOI 10.1002/sus2.70038): hard-segment content sweep from 17 % to 25 % in bio-based PUEs — enzymatic degradation primarily targets ester groups in *soft segments*; lower hard-segment content (17 %) → 82 % mass loss; higher hard-segment (25 %) → 36.8 % mass loss. **Confirms putting the cleavable bond in the soft phase resolves the tension.**
- **PCL-AD/PCL-DD elastomers** (Qin *Adv. Sci.* 2024, DOI 10.1002/adv.202400255): 100-cycle fatigue resistance plus good elastic recovery (residual strain 3–8 %) with PCL soft block, by tuning hard-segment cluster size via chain-extender choice.
- **PLA-PU with PEA soft segment + MDI/EG hard segment** (Gu et al., PMC11243797): 37.15 MPa stress, 820.8 % elongation, biodegradable >80 % in 6 months. Mechanically excellent and degradable, but uses MDI → reverts to Blind Spot 1; for our use case substitute H12MDI.

- **Hard-block degradable TPUs with carbonate motifs in hard block** (Stankus/Wagner cardiovascular work, ResearchGate 258669684): poly(hexamethylene carbonate) soft + HDI + cleavable bis(3-hydroxypropylene)carbonate chain extender → non-acidic degradation products, hard-block-dependent degradation rates.

The compositional region that balances both

Synthesizing the cited literature:

- **Soft segment:** PCL or polycarbonate diol, $M_n \sim 2,000$ g/mol, *containing the cleavable ester/carbonate linkages*.
- **Hard segment content:** 25–35 wt% (lower than commercial Lycra's ~30–40 % but still elastomeric); at 17 % degradation is too fast (82 % in 14 d) for use-phase durability.
- **Hard segment chemistry:** aliphatic isocyanate (HDI, H12MDI) + short-diamine chain extender (ethylenediamine, propylenediamine) — gives urea hydrogen-bonding network for elasticity *without* aromatic-amine carcinogenicity.
- **Cleavable bond placement:** in the *amorphous soft phase* (ester or carbamate adjacent to soft block), NOT in the crystalline hard domains.

Verdict: MANAGEABLE — published design window exists.

Defensible answer

"The trade-off is real and published. The honest version of the curve is: **at <17 wt% hard segment, the material degrades too fast for use-phase durability (82 % mass loss in 14 d, Xu *SusMat* 2026); at >40 wt% it doesn't degrade fast enough at end-of-life. The published design window is 25–35 wt% hard segment with PCL or polycarbonate soft block of $M_n \sim 2,000$ g/mol, aliphatic isocyanate (HDI/H12MDI) for the hard block, and the cleavable bond placed in the soft/amorphous phase rather than the crystalline hard domains.** This is the same compositional region Xu et al. *SusMat* 2026 and Qin et al. *Adv. Sci.* 2024 used to achieve elongation >400 %, fatigue resistance over 100 cycles, AND enzymatic degradability. Engine A's job is to search *within* this published window — not to discover that the window exists."

BLIND SPOT 6 — Engine A Reframe: Constraint-Guided Virtual Screening

Verified facts on existing tools

Public databases:

- **PoLyInfo (NIMS Japan)**: ~13,000 experimentally measured polymers with Tg, Tm, density, dielectric constant; includes "homopolymers, copolymers, polymer blends, composites" (<https://polymer.nims.go.jp/>). (Nd) (Nims)
- **PI1M (Ma & Luo 2020, DOI 10.1021/acs.jcim.0c00726)**: ~1 million virtual polymers generated from a generative model trained on PoLyInfo's 13,000 — used as a virtual-screening sandbox. (Nd)
- **Ramprasad group datasets** (Huan et al. *Sci. Data* 2016, DOI 10.1038/sdata.2016.12): ~1,000 polymers with computed properties for dielectric/energy applications.
- **OpenPoly (2025)**: benchmarking database for multi-property polymer prediction.
- **PolyID (NREL, Wilson et al. *Macromolecules* 2023, DOI 10.1021/acs.macromol.3c00994)**: graph neural network for bio-based homopolymer Tg prediction, MAE 19.8 °C test / 26.4 °C experimental — homopolymer-trained, no explicit segmented-PU support. (ResearchGate) (nih)

ML property-prediction tools applicable to PU-urea:

- **polyBERT** (Kuenneth & Ramprasad, *Nat. Commun.* 2023, DOI 10.1038/s41467-023-39868-6): 100M hypothetical PSMILES strings; downstream property dataset = "28,061 (~80%) homopolymer and 7456 (~20%) copolymer (total of 35,517) data points." Verbatim from the paper: "Our copolymer data points are for random copolymers... Alternating copolymers are treated as homopolymers with appropriately defined repeat units." **Does NOT support block/segmented copolymers** — only random and alternating. (PubMed Central) (PubMed Central)
- **polyBART** (Savit et al., EMNLP Findings 2025, arXiv:2506.04233): explicitly states verbatim: "polyBART is currently limited to homopolymers and does not yet support the representation or generation of copolymers, polymer blends, or polymer composites/formulations." (arXiv)
- **POLYT5** (Sahu et al., *npj Artificial Intelligence* 2026, DOI 10.1038/s44387-026-00087-1, arXiv:2510.18860): T5 encoder-decoder demonstrated for dielectric homopolymer design. No copolymer/segmented-PU disclosure. (arxiv) (ResearchGate)
- **PU-elastomer-specific ML** (*Chinese J. Polym. Sci.* 2022, DOI 10.1007/s10118-022-2838-6): Young's modulus, tensile strength, elongation at break of PUEs, R² 0.85–0.91 — *trained directly on PU elastomer data*, useful for Engine A.
- **PUR-GEN web server** (ResearchGate 381884352): generates PU oligomer libraries from isocyanate + alcohol building blocks for molecular docking — useful for monomer/chain-extender combinatorics.
- **BigSMILES** (ChemRxiv 2024, DOI 10.26434/chemrxiv-2024-bxxhh-v5): verbatim — "the ability of BigSMILES to succinctly describe complex polymer structures, such as copolymers, is a distinct advantage that SMILES lacks, which limits the current polymer ML workflows to primarily focusing on homopolymers." (ChemRxiv)

Feasibility of the reframed approach (audit's recommendation)

The audit's pivot is correct and tractable in 72 hours.

- None of polyBART, polyBERT, POLYT5, PolyID handles segmented block copolymers — polyBART says so verbatim. Using these models naïvely on segmented PU-urea SMILES will give garbage outputs.
- The viable 72-hour workflow is:
 1. **Enumerate** monomer combinations: aliphatic diisocyanates (HDI, H12MDI, IPDI) × soft-segment polyols (PCL, PHC, PTMEG, polyester diols) × chain extenders (ethylenediamine, propylenediamine, BDO) × hard-segment wt% (20, 25, 30, 35 %). ~500–2,000 candidates.
 2. **Build each as a hypothetical repeat unit** (with hard-segment-equivalent stoichiometry) and score with a PU-elastomer-specific Tg/modulus/elongation predictor (Springer 2022, DOI 10.1007/s10118-022-2838-6).
 3. **Filter** on constraints: Tg(soft) < -50 °C, Tg(hard) > 150 °C, elongation > 400 %, hard-segment 25–35 wt%, aliphatic isocyanate only, ester/carbamate bond placed in soft phase.
 4. **Emit** the top 10 candidates to Engine B (urethane substrate model) for enzyme active-site design against the soft-segment carbamate bond.
- This is NOT "AI-generative design" — it is **constraint-guided high-throughput virtual screening using off-the-shelf property predictors over a chemist-enumerated library**.

Verdict: NON-ISSUE if the team reframes Engine A honestly.

Defensible answer

"You are right that polyBART and polyBERT do not handle segmented block copolymers — polyBART explicitly says so in its own paper (Savit et al. 2025, arXiv:2506.04233). Engine A is not de novo generative design. Engine A is **constraint-guided virtual screening of the published polyurethane-urea design space**: we enumerate ~1,500 candidate compositions across aliphatic-diisocyanate × polyol × chain-extender × hard-segment-wt% combinations, score each with a PU-elastomer-specific Tg/modulus/elongation predictor (Springer *Chinese J. Polym. Sci.* 2022, DOI 10.1007/s10118-022-2838-6), filter against our property constraints, and emit a substrate model for Engine B. The 'AI' in Engine A is **predictive screening at scale, not generative novelty**. The novelty of REWEAR-FUSED is the *co-design loop* between Engine A (substrate constraints) and Engine B (enzyme design), not generative chemistry."

OVERALL ASSESSMENT

Ranking by danger to the pitch

1. **Most dangerous (FATAL if uncorrected): Blind Spot 1 (MDA/TDA toxicity).** A pediatric-apparel sustainability judge will immediately ask "what comes off the polymer when you cleave it?" If the team says "MDA from MDI hard segments," the pitch is dead — MDA is IARC 2B, REACH Annex XVII Appendix 8, OEKO-TEX Class I detection limit 20 mg/kg. The fix is mandatory and trivial in framing: constrain Engine A to aliphatic isocyanates from minute one.
2. **Second most dangerous: Blind Spot 2 (pH conflict).** UMG-SP2's verified pH-10 optimum is a published, citable, mechanistically explained fact that a biochemistry-literate judge will quote against the team. The team must lead with "we are not using wild-type UMG-SP2; Engine B engineers a near-neutral-pH Ser-His-Asp variant."
3. **Real but manageable: Blind Spot 5 (mechanics-vs-cleavability).** The published design window (PCL soft segment, 25–35 wt% hard, cleavable bond in amorphous phase, aliphatic isocyanate) is the team's defense.
4. **Real but manageable: Blind Spot 3 (cyclic fatigue/ML limits).** Honest concession that ML cannot predict fatigue; wet-lab validation required.
5. **Easily neutralized: Blind Spot 4 (CapEx/drop-in).** Hyosung Creora bio-based (1-year R&D-to-market, 23 % CO₂ reduction), LYCRA × QIRA (3-year, early-2025 launch), ROICA V550 (2-year-to-brand) are the precedents; license to incumbent, don't build a plant.
6. **Easily neutralized: Blind Spot 6 (Engine A reframe).** Reframe as "constraint-guided virtual screening," not generative design — this is more honest AND more credible.

Additional un-researched gap a sustainability judge will probe

Microplastic shedding during the use phase and end-of-life leaching of unreacted aliphatic diamines into wash water. The EU PFAS restriction proposal (ECHA Feb 2023) and the EU intentionally-added-microplastics ban (Commission Regulation 2023/2055) are the regulatory frame. Even an aliphatic-diamine-emitting elastane will release nanoparticles and free diamine during 200,000 laundering cycles. The team should pre-empt this by (a) citing the OECD 301-series biodegradability test framework, (b) committing to OECD 318 (Dispersion Stability of Nanomaterials) testing, and (c) framing enzymatic end-of-life recycling as *replacing* the microplastic-shedding-to-ocean fate with a controlled industrial hydrolysis fate. This is the sustainability-judge question the audit missed.

RECOMMENDATIONS (Staged, Concrete)

Stage 1 — Before the build starts (next 24 hours of prep):

- Lock Engine A's design space to **aliphatic isocyanates only** (HDI, H12MDI, IPDI). Hard-coded constraint in the screening pipeline.
- Lock the pH window for Engine B to **pH 7.0–8.5** with Ser-His-(Asp/Glu) catalytic triad as the design target, not Ser-Ser-Lys.
- Lock the soft segment to **PCL or polycarbonate**, ~2,000 g/mol, with the cleavable bond *in the soft segment*. Hard-segment range 25–35 wt%.

Stage 2 — During the 72 hour build:

- Engine A: implement constraint-guided enumeration + PU-elastomer-specific property prediction (Springer 2022 model). Output: 10 candidate PU-urea structures + carbamate substrate model.
- Engine B: design a Ser-His-Asp carbamate hydrolase using RFDiffusion / Chroma / LigandMPNN against the soft-segment carbamate substrate from Engine A.
- Engine C: build the Digital Recyclability Passport schema with explicit fields for **OEKO-TEX Class I expected pass, REACH Annex XVII Entry 43 + Entry 72 compliance, CPSIA 16 CFR Part 1610 flammability, and ISO 10993-5/-10 biocompatibility status**.

Stage 3 — Judge defense:

- Lead every "MDA?" question with the aliphatic-isocyanate slide.
- Lead every "pH?" question with the Ser-His-Asp re-engineering slide and the Branson 2023 verbatim quote.
- Lead every "manufacturing?" question with the Hyosung Creora / LYCRA-QIRA / ROICA precedent slide.
- Pre-empt the microplastic question with the controlled-hydrolysis-replaces-ocean-shedding framing.

Benchmarks that would change the recommendations

- If any Engine A candidate has **<25 wt% hard segment** AND still passes Tg/elongation filters → flag as too fragile for use-phase durability; widen filter.
- If Engine B's designed enzyme has a **predicted pH optimum >8.5** (e.g., AlphaFold-style pKa prediction of the catalytic base) → re-design with His instead of Lys.
- If any degradation product is on the **REACH Annex XVII Appendix 8 list, OEKO-TEX Class I list, or California Prop 65 list** → kill the candidate.

CAVEATS

- The numeric pH-7 residual activity of UMG-SP2 is not stated verbatim in Branson 2023, Bayer 2024, or Li 2025 — it must be read from the activity-pH curve in the Branson 2023 supplementary information. The team should reference the Branson 2023 verbatim "all enzymes were most active at pH 10" rather than invent a percentage.
- The 4-step de novo enzyme design pipeline (ProteinMPNN + RFDiffusion + Rosetta enzyme design + AlphaFold validation) has been demonstrated for new chemistries but *has not yet been demonstrated for a carbamate hydrolase against a novel aliphatic-PU substrate*. Expert judges will note this is an unproven combination.
- The published PCL/polycarbonate biodegradable PU elastomer literature is overwhelmingly from biomedical-device contexts (vascular grafts, scaffolds), not apparel. Wash-durability in 200,000-cycle laundering has not been demonstrated for any of the cited compositions.
- LYCRA × QIRA preserved equivalent function only because they replaced *one* feedstock with a chemically identical bio-version. The REWEAR-FUSED team is proposing a *different chemistry* (aliphatic isocyanate, cleavable soft segment) — closer to ROICA V550's 2-year-to-brand timeline plus several years of prior R&D than to Hyosung Creora's 1-year drop-in timeline. The defensible commercialization horizon is **2030–2032, not 2027–2028**.
- All toxicity claims for aliphatic-isocyanate-derived diamines (1,6-hexamethylenediamine, 4,4'-diaminodicyclohexylmethane, IPDA) are based on their *absence* from the IARC 2A/2B carcinogen list and from REACH Annex XVII Appendix 8 — not on positive demonstration of safety. ISO 10993 testing on actual hydrolysis extractables is required for any Class I infant claim.
- Spandex market context: China alone produced **939,000 tons in 2023 (+19.5 % YoY) with installed capacity of 1.2395 million tons/year** (CCFGroup via market.us, Dec 2024); a novel chemistry licensed at even 1 % market penetration would be commercially significant. (Market.us)